NewsLens: Explainable News Classification

Problem: a topic label alone does not tell a user when to trust or review a result.

Pipeline: English article → three class scores → relative terms → accept or manual review.

Yi Tan (2312260). The live demo uses a local lexical model.

System and Evaluation Protocol

11,519 labeled articles; shared train / validation / test counts: 9,215 / 1,152 / 1,152.

Seed 42 and saved row identifiers; vocabulary, IDF, and NYT embeddings use training text only.

Original components: calibration, threshold selection, additive explanation, deletion check, interface, and tests.

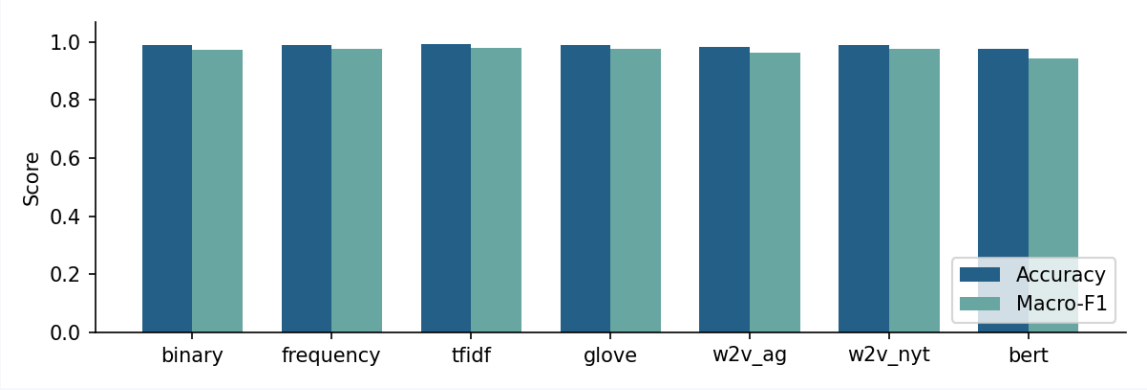
Measured Classification Baselines

Binary BoW: 98.78% accuracy, 97.25% macro-F1.

Word Frequency: 99.05% accuracy, 97.74% macro-F1.

TF-IDF: 99.13% accuracy, 98.03% macro-F1.

The strong linear baseline motivates separate tests of confidence and explanation quality.



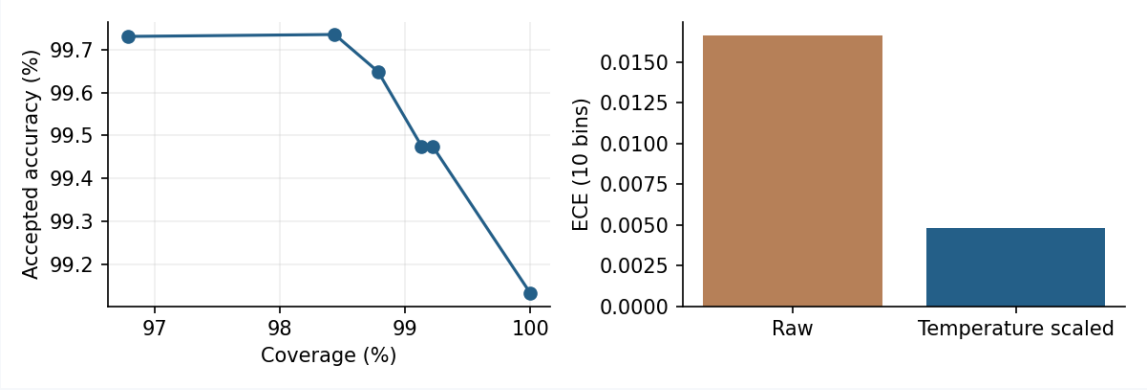
Calibration and Explanation Tests

Test ECE before / after scaling: 0.0167 / 0.0048.

Selected threshold: 98.44% coverage; 99.74% accuracy among accepted articles.

Top-five feature deletion changes 7.00% of predictions; random deletion changes 0.00%.

Contributions explain linear scores; they are not causal facts about the news.



Challenges and Next Steps

72 exact duplicate texts; eight test texts appear in training.

Every test article exceeds the BERT 64-WordPiece input limit.

Complete required neural and GloVe records, then refresh the formal milestone.

The individual contribution and Codex assistance are disclosed separately.